

Distribution of a sum of random variables

The random variables X and Y have the following joint distribution:

X	$Y=1$	2	3	4	
1	0.01	0.04	0.02	0.03	
2	0.04	0.05	0.08	0.03	
3	0.07	0.10	0.08	0.05	
4	0.08	0.11	0.12	0.09	

- Add the marginal distributions for X and Y to the table.
- The sum $X+Y$ is a new random variable. What are its possible values?
- Compute the distribution of $X+Y$. (Give the probability for each possible value of $X+Y$.)
- Compute $E[X]$, $E[Y]$, and $E[X+Y]$. How are these values related?